

SDR Thermal Stabilization

I have always had the SDR in a remote box rather than running a long (25m) coax. In the past, the SDR & USB Extender box had been on a shaded screen porch and was (by chance) somewhat temperature stabilized .

Over the past several months I have been building and refining highly portable Disk Yagi antennas and for convenience, installed all electronics at the antenna.

What I did not anticipate was that an AirSpy mini SDR is quite sensitive to temperature changes . Not gain, but a significant change in spectral response : the filter shape changes .

Moving forward , I've finally stabilized the SDR by installing it within a small insulated box. June 9th was the first test of the TEC cooled , Resistive heated thermal control system

I've never had such stable data . An example is a spectrum plot set (135 ea) over 11 hours where the air temp changed from a morning low of 74F to a late afternoon high of 102F in the sunlit area around the SDR box ...

The large shift was due to the Sun transit within ~ 4 deg of the Declination +20 drift scan. It provides a nice separation of the 5 minute duration spectra scans.

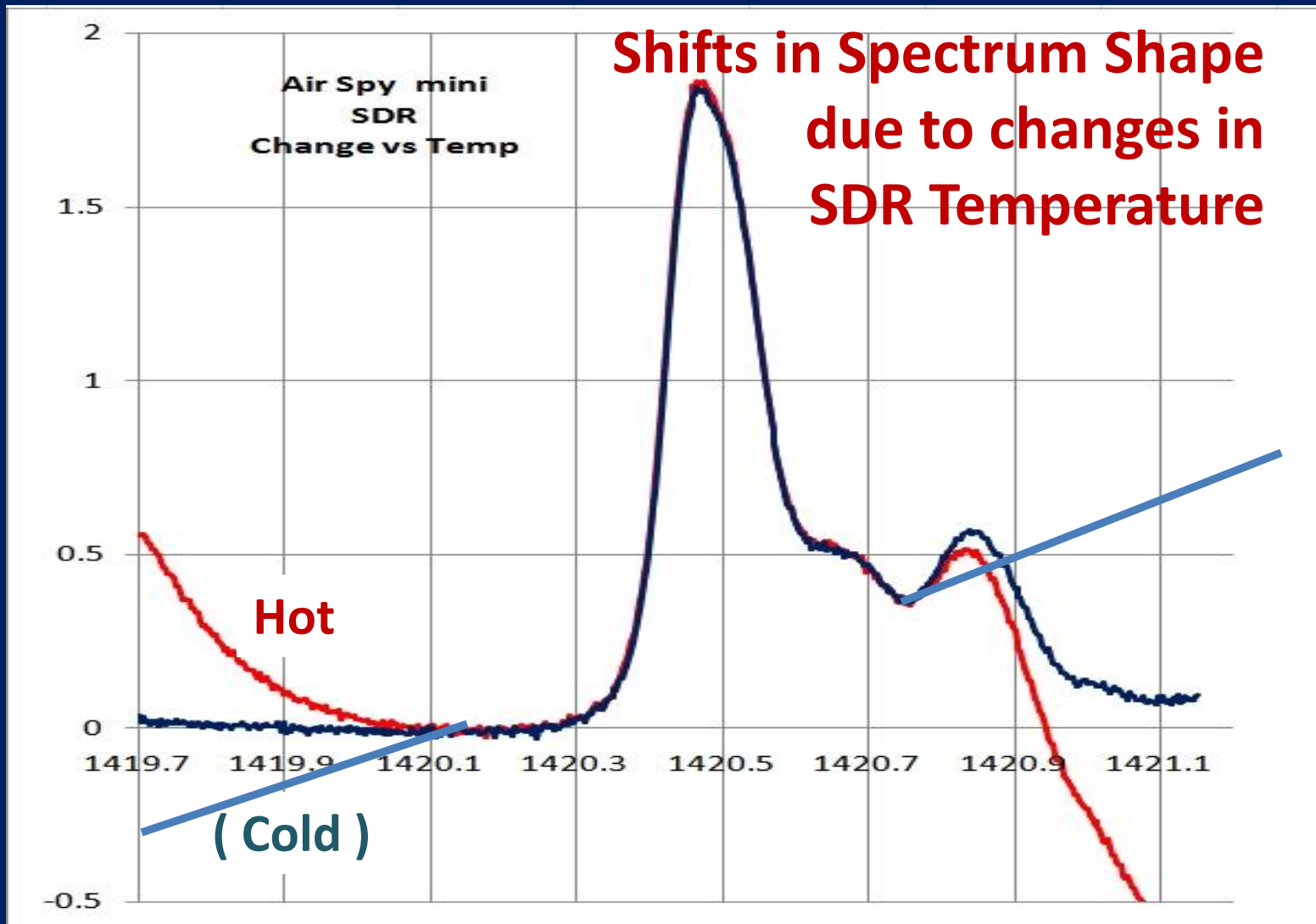
youtubevideo : **SDR Thermal Stabilization System @ + 26 minutes**

https://www.youtube.com/watch?v=fEpjte5dZ_k

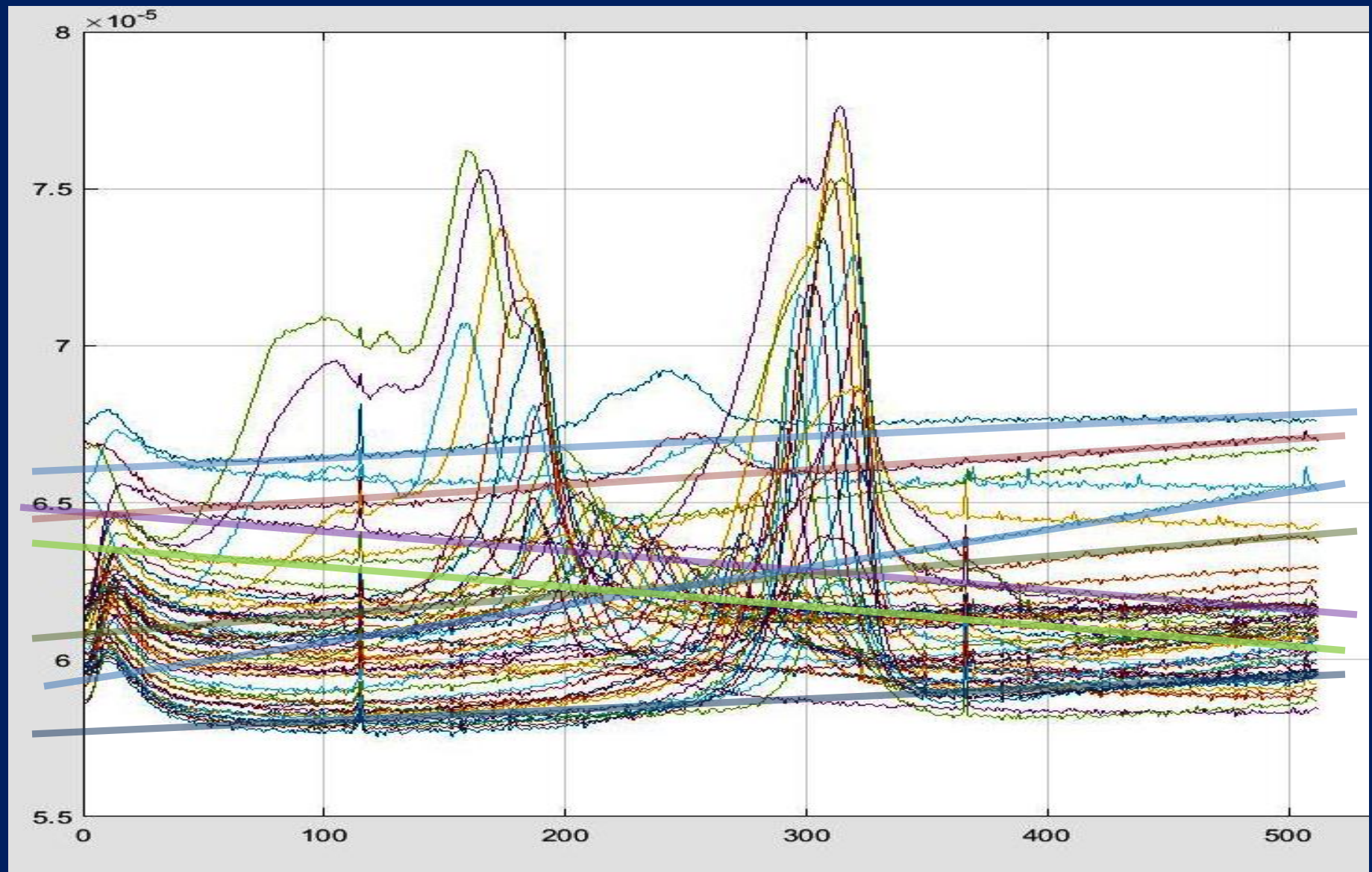


bapettitjr June '24

The Problem



The Problem = Example 24hr (every 5th spectra)



Fast Forward Three Months and a Dozen Variations : **The Solution**

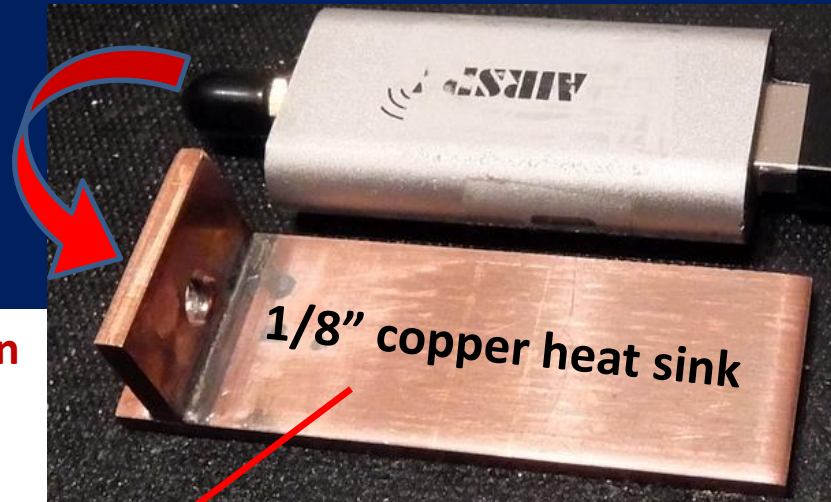
**Thermally Stabilize the SDR
within a temperature controlled box**



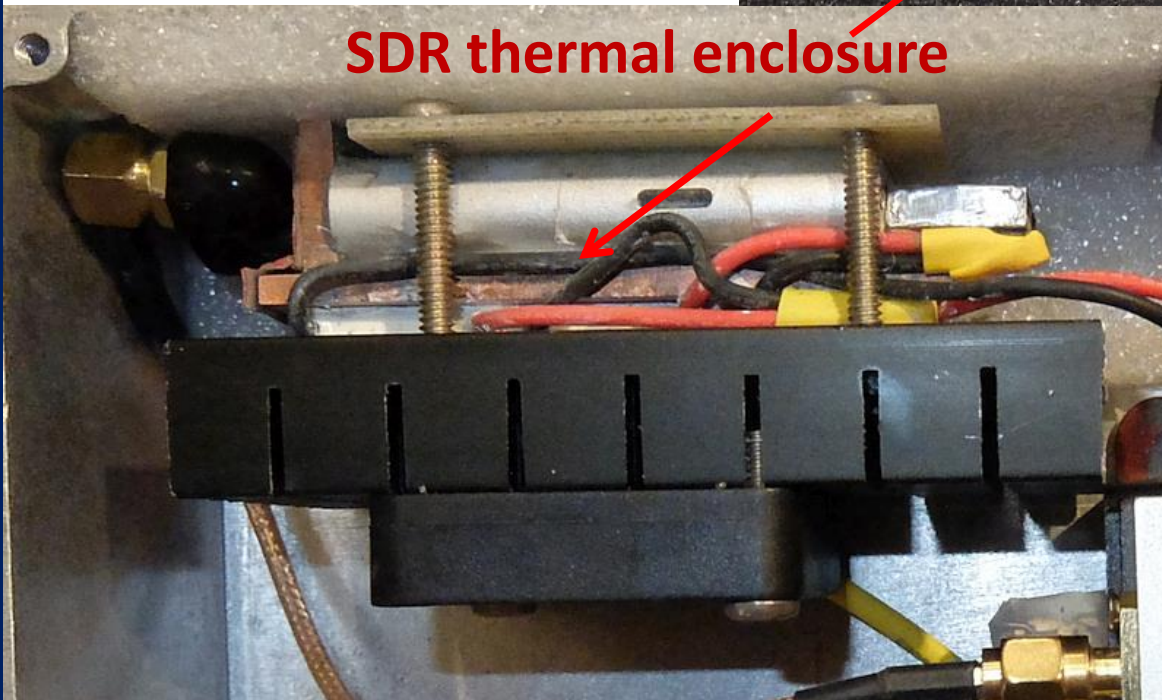
The Implementation

Primary Heat Transfer Path
for PC board is via the
SMA Connector Shield

Thermally Stabilize the SDR within
a temperature controlled box
(foam enclosure)

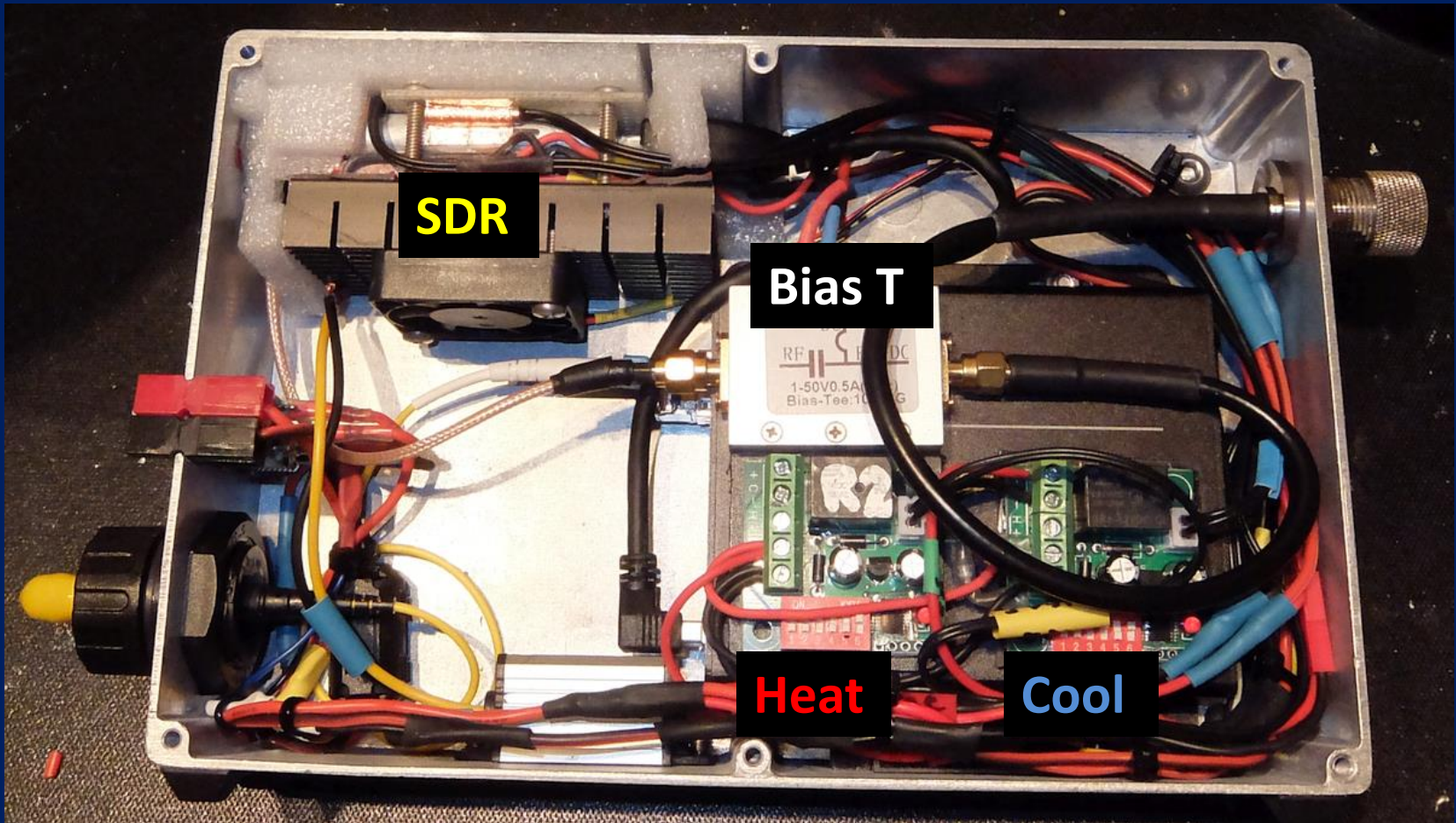


SDR thermal enclosure



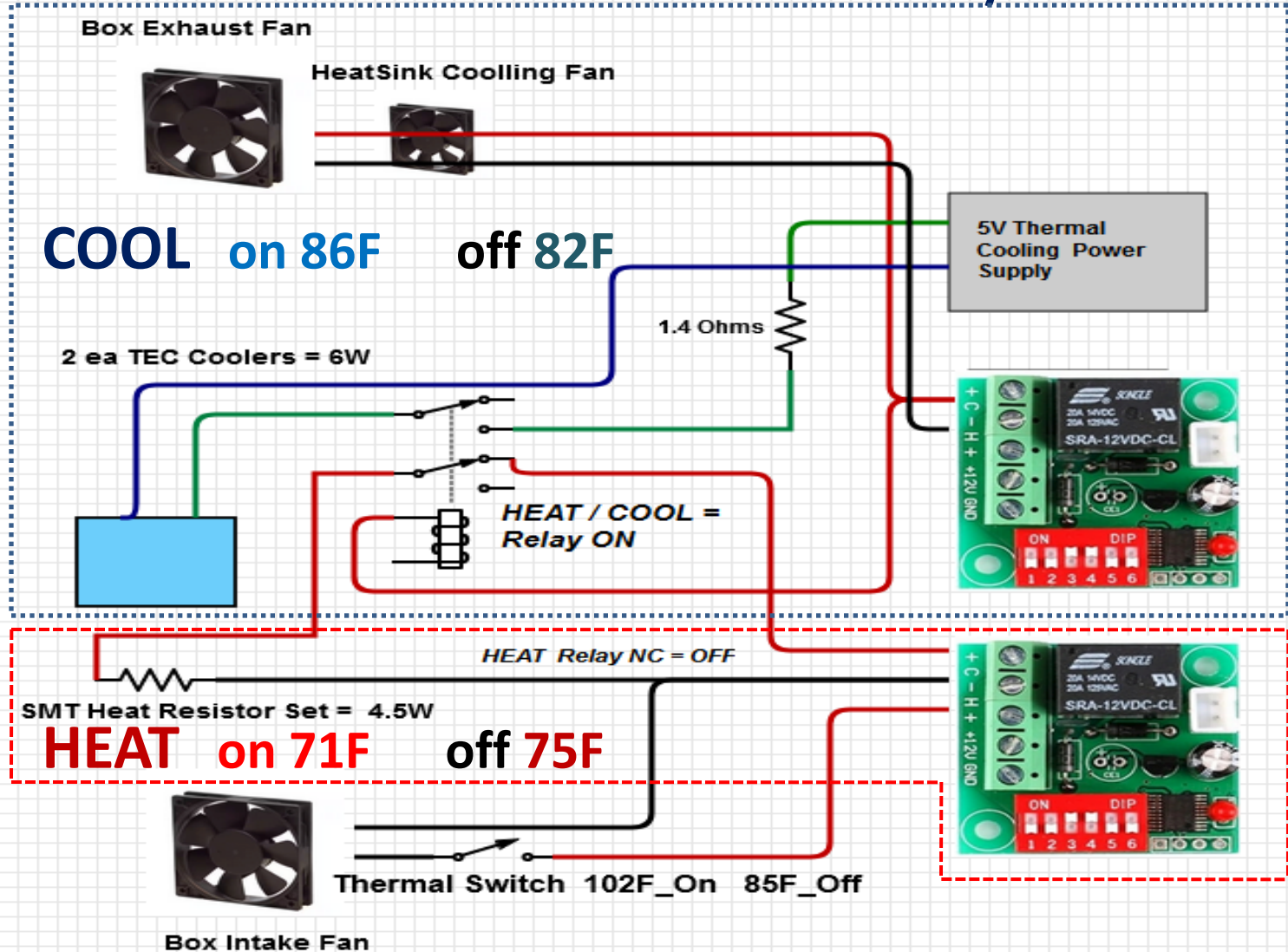
The Implementation

SDR Thermal Stabilization Control System

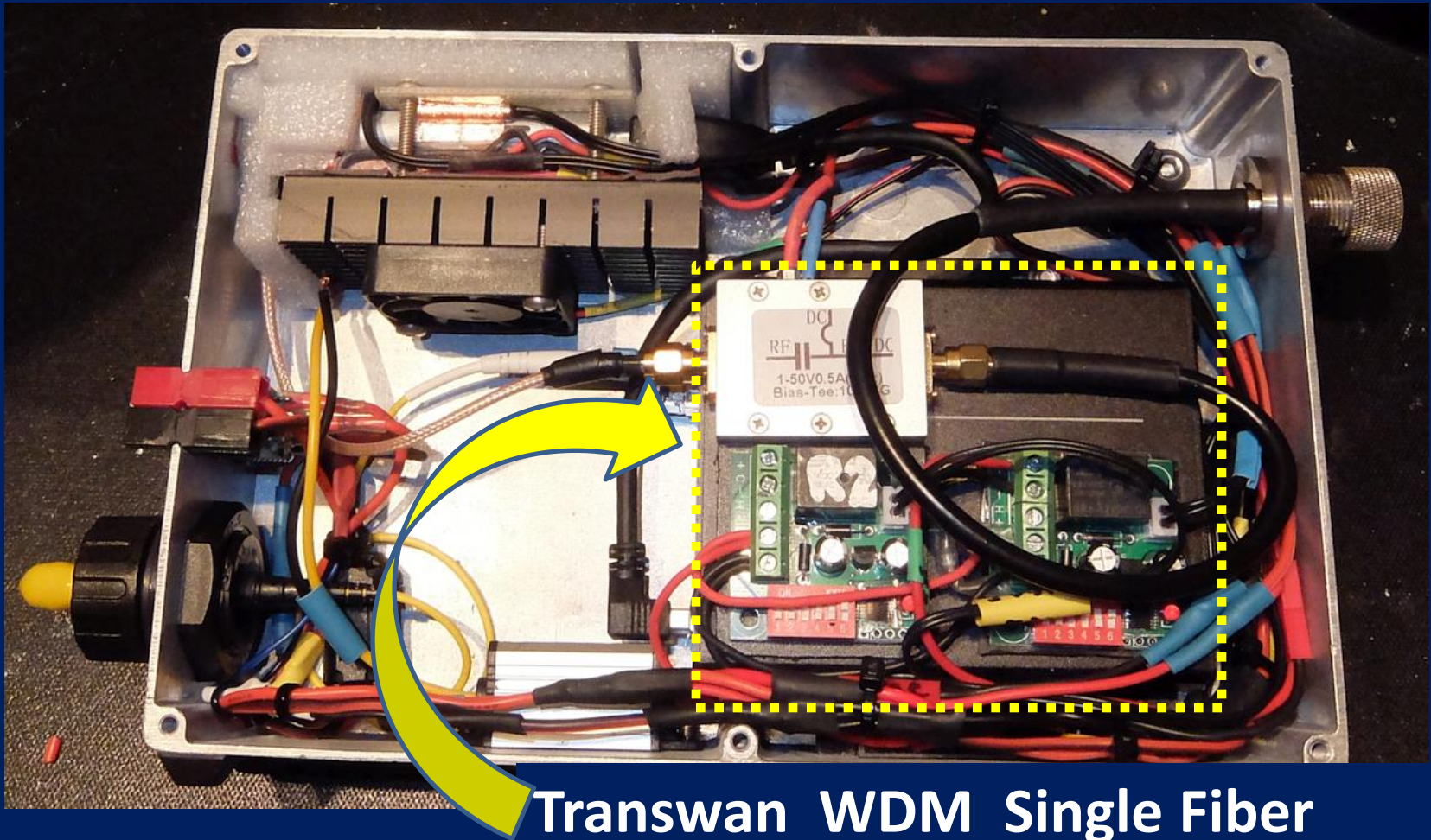


The Implementation

SDR Thermal Stabilization Control System



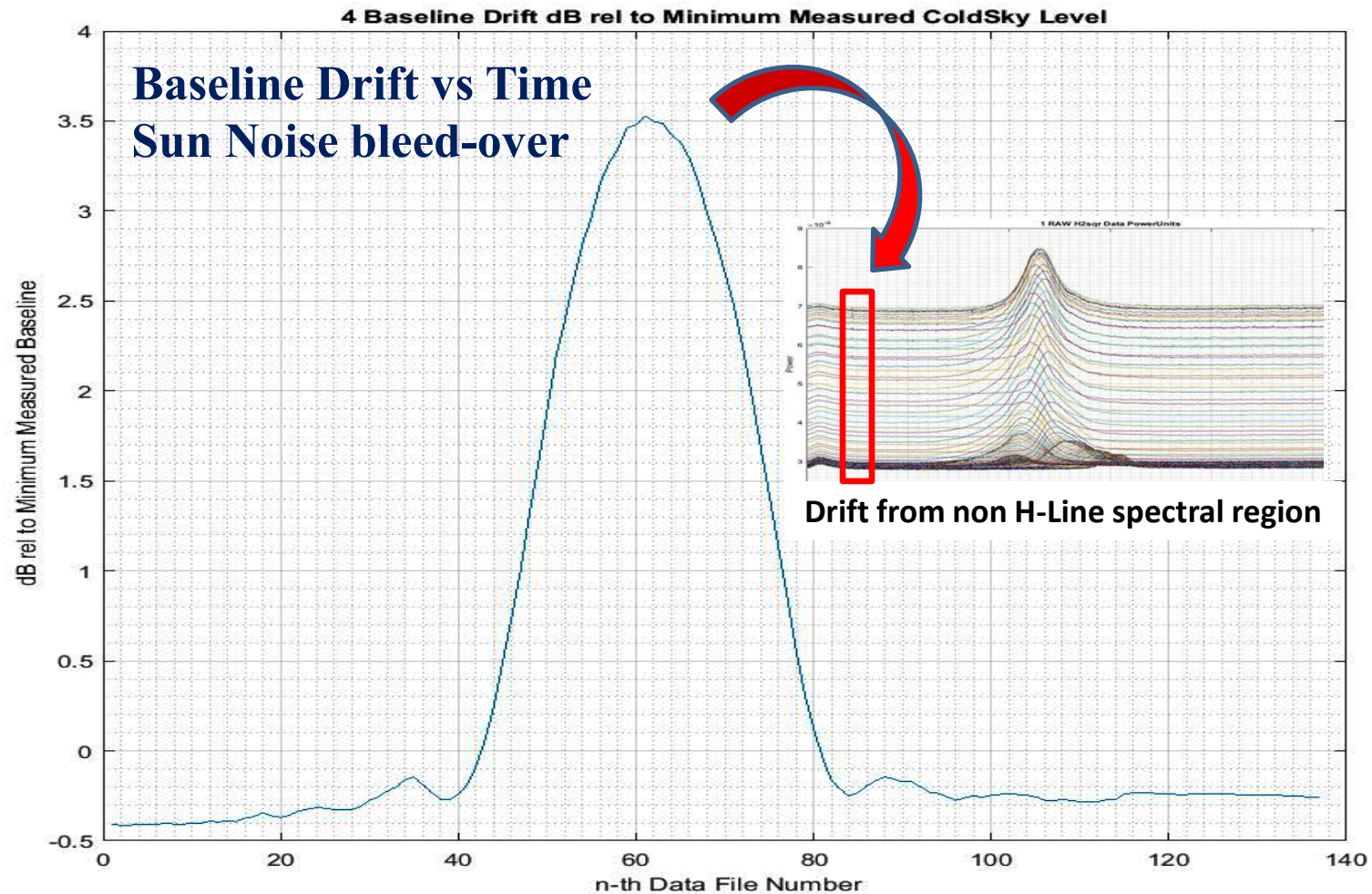
The SDR Digital Data Link



**Transwan WDM Single Fiber
USB over Fiber Optic Transceiver**

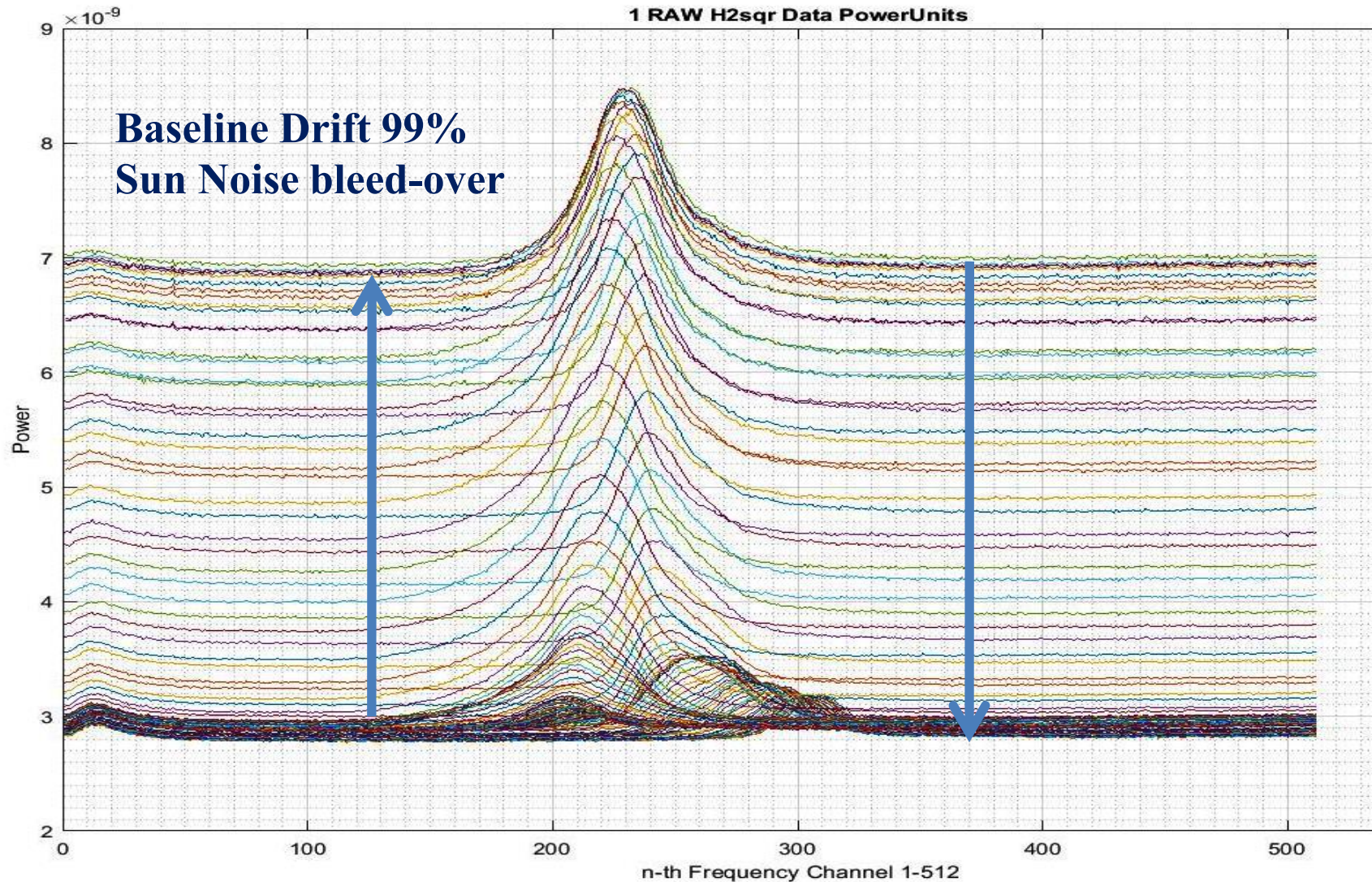
The Results

Baseline Drift = Sun Noise Bleed-Over



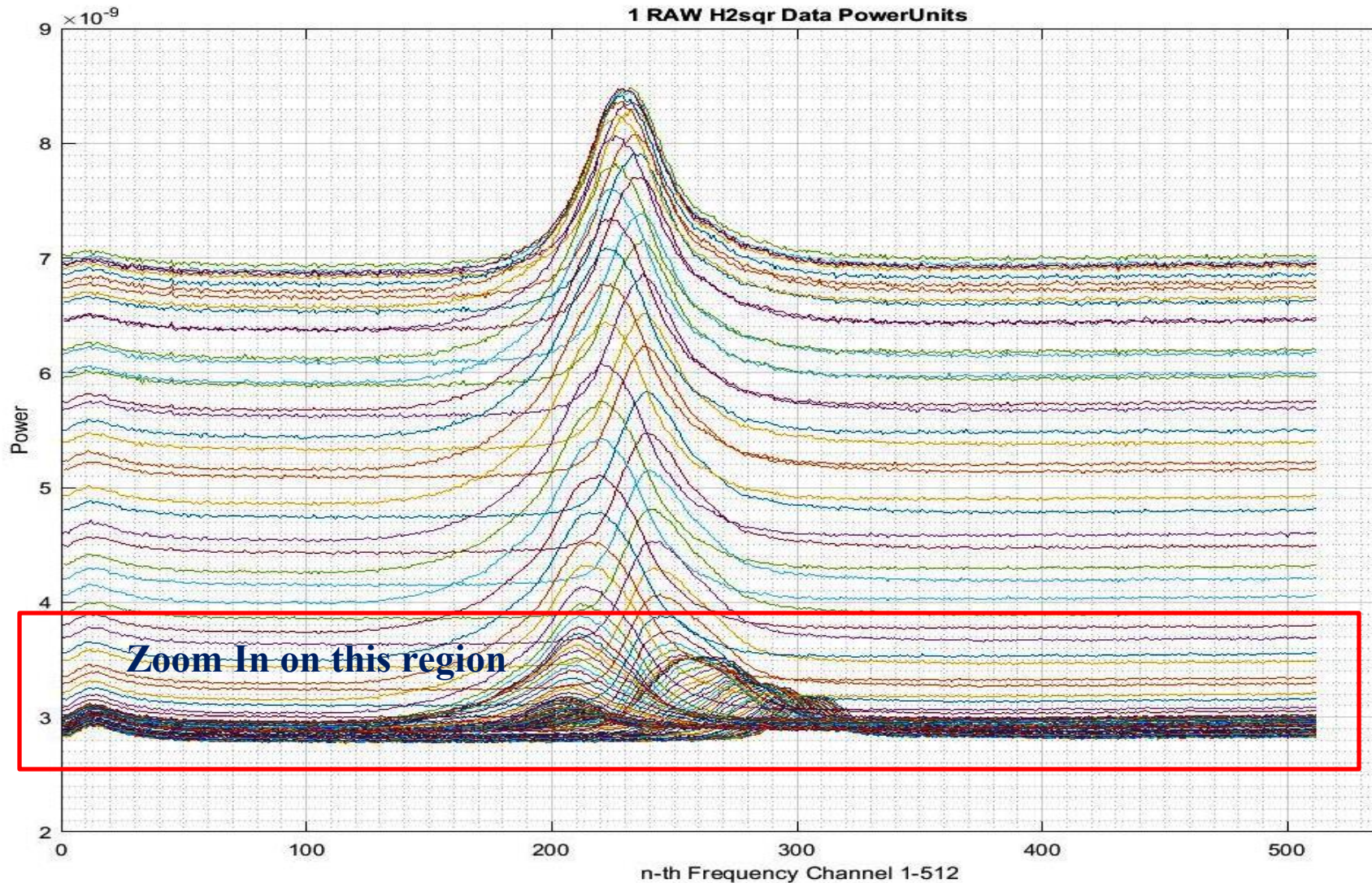
The Results

uncorrected raw power spectra samples



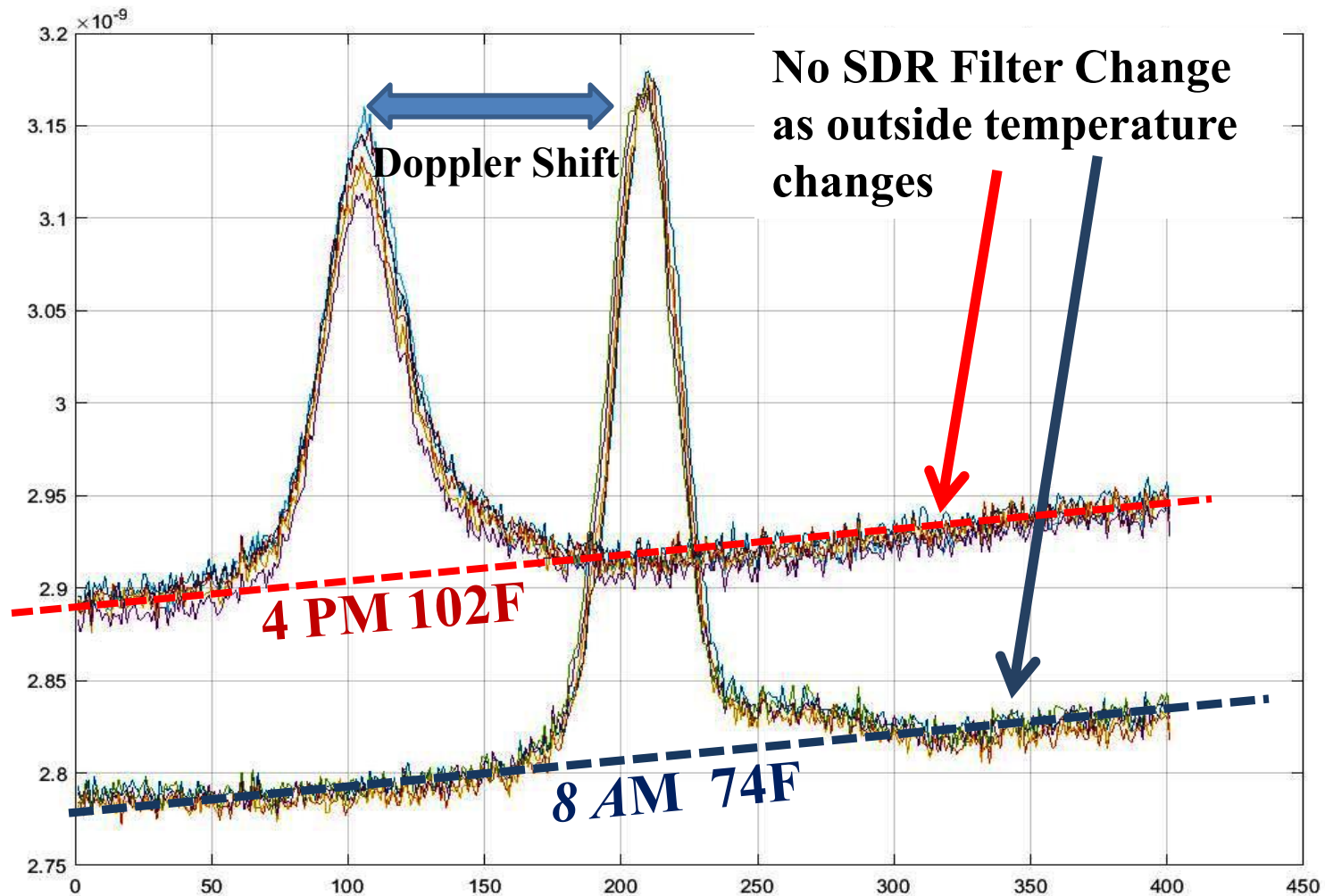
The Results

uncorrected raw power spectra samples



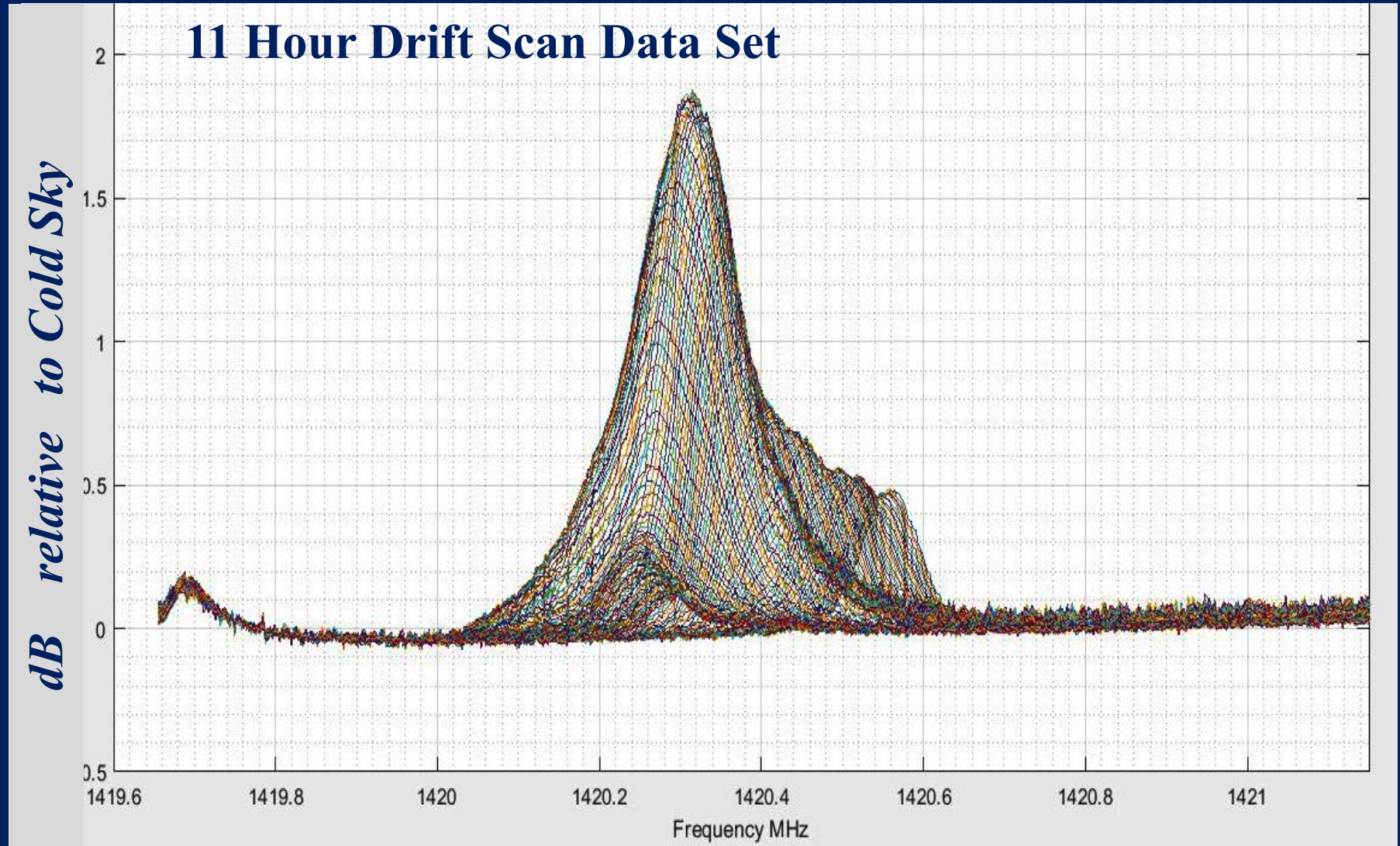
The Results

Data Samples 8AM @ 75F & 4PM @ 102F



The Results

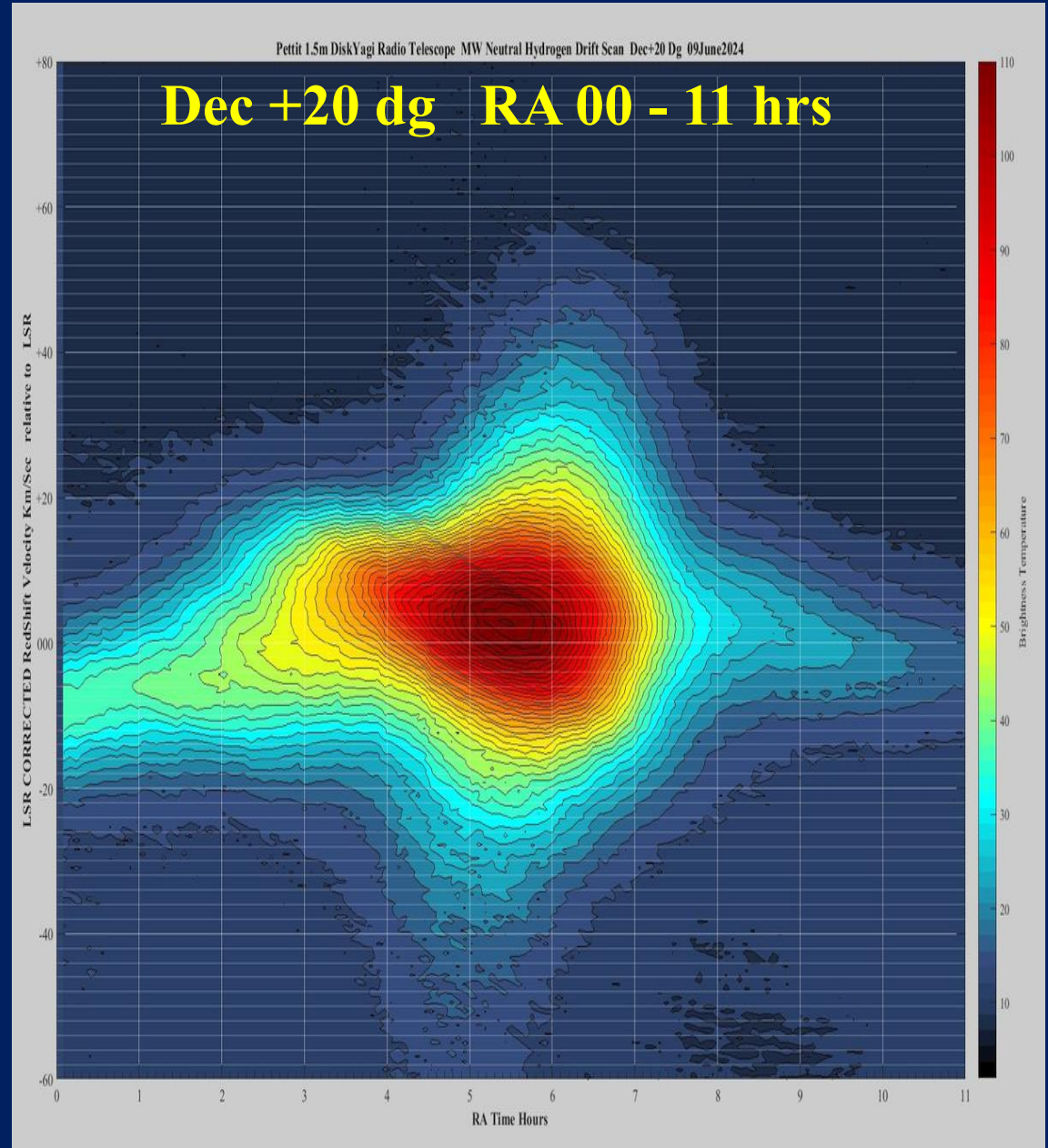
Baseline Shift Corrected & re-scaled in dB relative to Cold Sky



The System



The Analysis



Matlab H-Line Processing Software

uncorrected raw
spectra samples

normalized baseline
shift dB vs sample

baseline shift
corrected data
dB re Cold Sky

VLSR
correction curve

Spectra Set
pre/post VLSR correction

